Cong Hoan Ngo

Data Engineer

Hanoi, Vietnam

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Building data pipelines and analytics systems for business decisions.

Summary

Data Engineer with 7+ years in software engineering and a data engineering focus since 2023, working for US and Singapore businesses in B2B distribution, healthcare, and pharma. Builds and runs ELT platforms with Apache Airflow, Airbyte, and dbt on Snowflake and BigQuery: 150+ Airflow DAGs and 200+ scheduled Metabase reports for a US distribution business, and reconciled healthcare and pharma data behind recurring reporting. Also builds LLM-powered data pipelines and manages infrastructure as code (OpenTofu) with workloads on Kubernetes.

Skills

Programming: Python, SQL, Java, JavaScript/TypeScript

Data Engineering: ETL/ELT, Apache Airflow, Airbyte, dbt, data modeling, data quality, web scraping, Selenium, Crawl4AI

Databases & Warehousing: Snowflake, BigQuery, PostgreSQL, MySQL, SAP HANA, Redis

Cloud & Infrastructure: AWS, GCP, Azure, OpenTofu/Terraform, Docker, Kubernetes, Helm, CI/CD

AI & LLM: OpenAI & Anthropic APIs, prompt engineering, LLM evaluation, embeddings & pgvector

Software Engineering: REST APIs, FastAPI, Flask, Spring Boot, React, Angular

BI & Analytics: Metabase, Tableau

Languages: English - C2 Proficient (CEFR)

Work Experience

Pod Co.

Data Engineer · Apr 2025 – Present · Remote · US-based company

- Build and operate 150+ Airflow DAGs and Airbyte pipelines into Snowflake for Wholesale, B2B, and Distribution.
- Develop dbt models and tests on Snowflake behind 200+ scheduled Metabase reports for business, finance, and sales.
- Built LLM-powered discovery on scraped social media: 100+ potential brands per week, over 30% became partnerships.
- Build internal web apps and tools for the team with React, such as a brand detection tool.
- Made dashboards faster to refresh and less prone to breaking, holding cost flat and cutting it 20–30% on some workloads.
- Prototyped an LLM text-to-SQL analytics assistant on Snowflake with query guardrails and LLM evaluation.
- Maintain infrastructure as code with OpenTofu (Terraform) and monitor data workloads on Kubernetes (Helm).

VMO Holdings Technology JSC

Data Engineer · Dec 2023 – Apr 2025 · Vietnam

- Migrated legacy cron pipelines to Apache Airflow, improving scheduling, monitoring, and maintainability.
- Built Python ETL workflows for multi-source healthcare and pharma data in a data lake and warehouse (Azure, BigQuery).
- Supported critical data source migrations while preserving data integrity and downstream compatibility.
- Improved pipeline reliability and reconciliation, enabling analytics delivery for 200+ recurring reports.
- Investigated data discrepancies with analytics teams and delivered Tableau dashboards for business users.

Maxflow R&D Co. Ltd.

Software Engineer / Full-stack Developer · Jan 2021 – Dec 2023 · Vietnam

- Developed core features and REST APIs for an industrial simulation and analytics platform with configurable workflows.
- Moved Python Flask services to serverless AWS (Lambda, API Gateway, S3), cutting infrastructure cost by about 25%.
- Implemented simulation modules for distance, energy, wear, and lifecycle cost across 100+ modeled scenarios.
- Built data preparation and feature extraction pipelines on Snowflake for predictive maintenance experiments.

FPT Information System Corp.

Back-end Developer · Dec 2019 – Dec 2020 · Vietnam

- Built Java (Spring Boot) backend services for a national public procurement platform in a 5-engineer team.
- Contributed to Angular-based front-end improvements, reducing average page load time by approximately 20%.

House3D Co. Ltd.

Web Developer · Feb 2019 – Dec 2019 · Vietnam

- Developed and maintained customer-facing web features for an interior design technology platform.

Framgia Inc. / Sun* Inc.

Web Developer Intern · Jun 2018 – Sep 2018 · Vietnam

- Built a Ruby on Rails e-commerce project in a 3-member team with PayPal, BaoKim, and NganLuong payments.

Projects

OnyxData DataDNA Analytics Portfolio

Personal Project · Python, Polars, DuckDB, Malloy

- Monthly challenges since May 2025: raw data to a parquet star schema (Polars, DuckDB) and a Malloy semantic model.
- Built an eight-gate validation pipeline that recomputes every published number before each release.

Discode: AI Coding Assistants in Discord

Personal Project · Python, PostgreSQL, Redis, Docker

- Built a Discord bot platform that runs AI coding CLIs (Claude, Codex, OpenCode) as stateful, thread-based chat sessions.
- Designed a multi-process bot/runner/dispatcher on PostgreSQL and Redis with an outbox and idempotent job claims.

Education

VNU – Hanoi University of Science

Bachelor of Science in Computer and Information Science · 2016 – 2021 · Vietnam

- Coursework focus: Software Engineering, Database Systems, Data Structures and Algorithms.

Certifications

IBM Data Architecture · May 2025

Google Associate Cloud Engineer · Dec 2024

Google Advanced Data Analytics · Nov 2024

IBM Data Engineering · Jul 2024

Microsoft Azure Data Engineering Associate · Jun 2024

AWS Cloud Solutions Architect · Jun 2024

Honors & Awards

Employee of the Year (VMO Year-End Award, VMO Holdings) · 2024